DATA QUALITY & GOVERNANCE

Customer Data Quality & Governance Audit

A Power BI portfolio project for Safeguard Insurance Limited

PROJECT PURPOSE
Identify customer data quality problems, show where records need review, and recommend practical actions that can help an insurance business maintain more reliable customer information.

Project detail	Description
Business scenario	Safeguard Insurance Limited (simulated case study)
Project type	Customer data quality assessment and governance reporting
Tools	Microsoft Excel, Power Query, Power BI Desktop, DAX
Dataset	1,250 synthetic customer records with intentional data quality issues
Main outputs	Executive Overview dashboard and Detailed Quality Analysis dashboard

Prepared as a portfolio case study

1. Executive Summary

This project presents a customer data quality audit for Safeguard Insurance Limited, a simulated insurance company. The company's customer records contain examples of missing information, invalid values, inconsistent categories, and records that may refer to the same customer. These issues can make reporting less reliable and make it harder for staff to contact customers or maintain accurate records.

Power Query was used to prepare the dataset and create flags for selected data quality issues. DAX measures were then used to summarise the records and calculate key indicators. Two Power BI report pages were built: an Executive Overview for headline indicators and a Detailed Quality Analysis page for specific issues and suggested corrective actions.

CURRENT DASHBOARD SNAPSHOT

The report currently displays 1,250 total records, 319 records requiring review, 931 records passing the defined checks, a 25.5% review rate, a 74.5% passing rate, and 60 potential duplicate records flagged. These are the current displayed results and should be treated as provisional until the underlying flags and duplicate logic have completed final validation.

2. Business Background and Problem Statement

Insurance companies rely on customer information to manage policies, communicate with customers, prepare reports, and support day-to-day operations. If the information is incomplete, incorrectly formatted, inconsistent, or duplicated, staff may spend extra time investigating records and reports may give a misleading picture of the customer base.

In this simulated scenario, Safeguard Insurance Limited needs a practical way to understand the condition of its customer data. The company needs to know which types of issues appear, how many records have been flagged, which records require follow-up, and what actions could reduce these problems.

Problem statement

There is no single, easy-to-read view of the main quality issues in the customer dataset. Without clear checks and monitoring, missing or invalid values and potential duplicate records may remain unnoticed or be handled inconsistently.

Business questions

- How many customer records are included in the dataset?
- How many records have been flagged for review, and what share of the dataset do they represent?
- Which missing or invalid values occur most often?
- How many records have been flagged as potential duplicates?
- What practical actions should the business take to investigate and prevent these issues?

3. Project Objectives and Scope

The project was designed to:

- Prepare a customer dataset for analysis while retaining a raw-data query for reference.
- Identify selected missing, invalid, and potentially duplicated records using rule-based flags.
- Create DAX measures to summarise the dataset and the flagged issues.
- Present the results in two Power BI report pages.
- Provide corrective-action recommendations that a data or operations team could use as a starting point.

Scope note: This is a portfolio demonstration using synthetic data. It is not an audit of a real insurer, and it does not confirm that any real customer information or company systems were examined.

4. Dataset Description

Item	Description
Dataset size	1,250 customer records (as currently shown in the dashboard)
Data source	Excel file imported into Power BI Desktop
Example fields	Customer details, contact information, city, policy type/status, signup date, date of birth, premium values, and matching fields used in duplicate analysis
Designed issues	Missing email/phone/name/city, invalid email/phone, invalid dates, invalid premium values, inconsistent categories, and potential duplicates
Use of data	Training and portfolio analysis only; records are synthetic

The report focuses on data quality rather than making business claims about policy performance, profitability, or customer behaviour. A flagged value indicates that a rule has identified something for review; it does not by itself prove fraud, customer error, or a confirmed duplicate.

5. Tools and Technologies

Tool	How it was used
Microsoft Excel	Held the source customer dataset.
Power Query	Prepared data, applied transformations, created quality flags, and supported the duplicate-count merge.
Power BI Desktop	Hosted the data model, report pages, visuals, and manual refresh workflow.
DAX	Created measures for record counts, review rates, and individual issue counts.

6. Data Preparation and Quality Checks

The dataset was imported from Excel into Power BI. A raw query named CustomerData_Raw was retained, and a main query named CustomerData_WithDuplicateCounts was used for analysis. The main query includes rule-based flag columns and a duplicate-count step.

Quality flags in the model

Flag	Purpose
Missing Email Flag	Marks records where the email field is missing or blank.
Missing Phone Flag	Marks records where the phone field is missing or blank.
Missing Name Flag	Marks records where the customer name is missing.
Missing City Flag	Marks records where city information is missing.
Invalid Email Flag	Marks email values that fail the project's validation rule.
Invalid Phone Flag	Marks phone values that fail the project's validation rule.

Invalid Premium Flag	Marks premium values that fail the project's rule.
Signup Date Invalid Flag	Marks signup dates identified as invalid.
Date of Birth Invalid Flag	Marks dates of birth identified as invalid.
Potential Duplicate Flag	Marks records identified by the duplicate-detection approach for further checking.
Needs Review Flag	Marks records that meet the project's criteria for review.

Preparation approach

- Keep a raw query available so that transformed results can be compared with the source.
- Use Power Query steps to prepare the fields used by the report and create issue flags.
- Treat blank and null values deliberately, rather than assuming that every empty field is a valid value.
- Use matching keys and a duplicate-count query to support potential duplicate detection.
- Retain flagged records for investigation instead of automatically deleting or changing customer information.

7. Data Model and DAX Measures

DAX measures calculate the totals and rates displayed in the report. The main measures use the CustomerData_WithDuplicateCounts table. The Needs Review Flag is compared with the text value "1" in the reported formula, so the flag's data type and values must remain consistent.

Measure	Purpose
Total Records	Counts records in the analysis table.
Records Requiring Review	Counts records where Needs Review Flag equals "1".
Records Passing Checks	Counts records classified as passing the defined checks.
Review Rate (%)	Calculates records requiring review as a share of total records.
Records Passing Checks (%)	Calculates records passing checks as a share of total records.
Missing Emails	Counts records flagged for missing email addresses.
Missing Phone Numbers	Counts records flagged for missing phone numbers.
Invalid Emails	Counts records flagged for invalid email values.
Potential Duplicate Records	Counts records flagged by the duplicate-detection logic.
Missing Names / Missing Cities	Counts records flagged for missing name or city information.
Invalid Phone Numbers	Counts records flagged for invalid phone values.
Invalid Signup Dates / Invalid Date of Birth	Counts records flagged for invalid date values.
Invalid Premium Values	Counts records flagged for invalid premium values.

Example DAX measure

```
Records Requiring Review =
CALCULATE(
    COUNTROWS('CustomerData_WithDuplicateCounts'),
    'CustomerData_WithDuplicateCounts'[Needs Review Flag] = "1"
)
```

This measure counts rows in the main analysis table where the Needs Review Flag is set to "1". The flag's logic determines which records are included, so its definition is just as important as the measure itself.

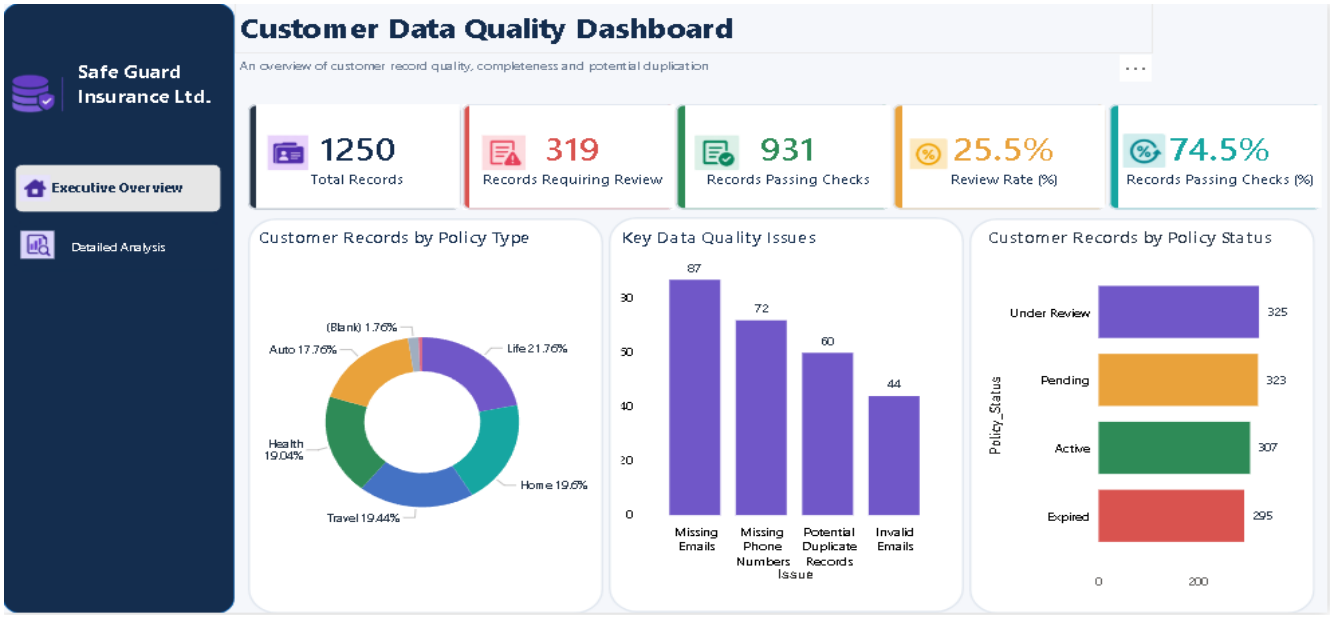
The report also uses helper tables such as DataQualityIssues and InvalidValueCategories to display issue categories through SWITCH-based measures. These tables help a single visual show counts from several different measures.

8. Dashboard Design

Page 1: Executive Overview

The Executive Overview is designed for a quick review of the dataset. It shows the total number of records, the number requiring review, the number passing checks, and the related percentages. It also summarises key quality issues and provides views of records by policy type and policy status.

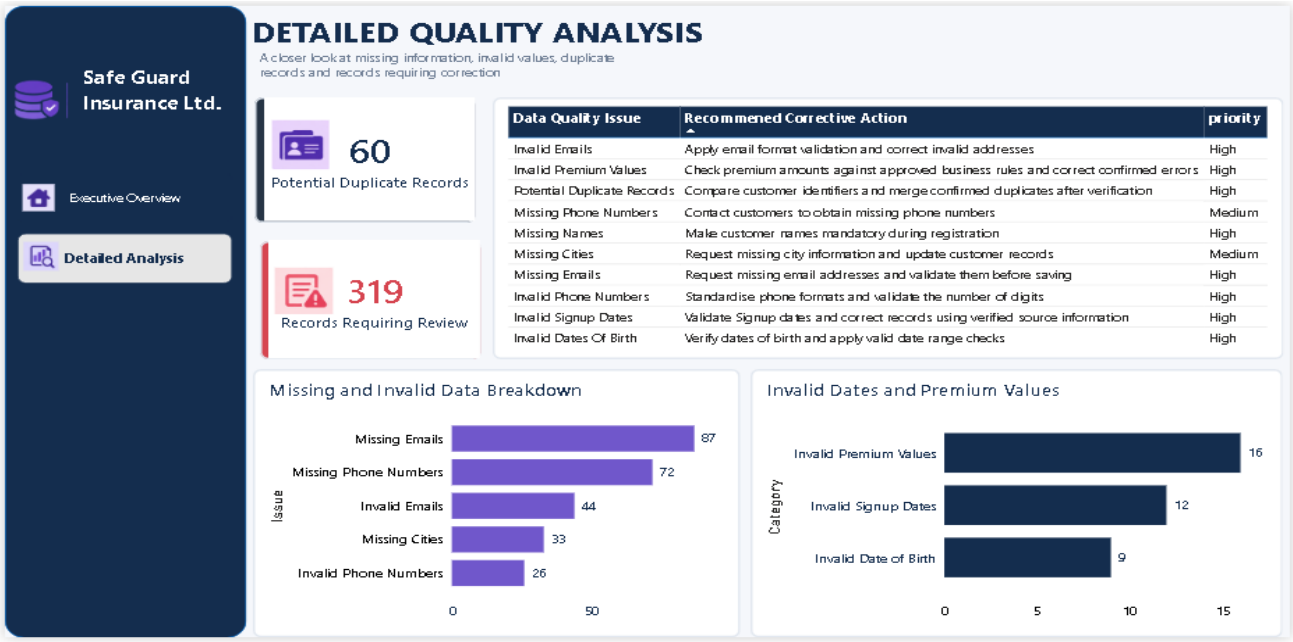
- Total Records, Records Requiring Review, and Records Passing Checks cards.
- Review Rate and Passing Checks percentage cards.
- Key Data Quality Issues chart.
- Customer Records by Policy Type visual.
- Customer Records by Policy Status visual.



Page 2: Detailed Quality Analysis

The Detailed Quality Analysis page gives more detail about specific issues. It includes counts for missing and invalid values, invalid date and premium values, potential duplicates, records requiring review, and a corrective-actions table.

- Missing and Invalid Data Breakdown chart.
- Invalid Dates and Premium Values chart.
- Potential Duplicate Records card.
- Records Requiring Review card.
- Corrective Actions table with issue, recommended action, and priority.



Testing completed so far: page navigation works; selecting a policy type in the Executive Overview changes related visuals; the Detailed Quality Analysis cards and corrective-action table display as expected. Selecting a bar in the detailed breakdown did not change other visuals in the current test. This behaviour may be acceptable for the current design, but it should be documented as a deliberate design choice or adjusted if cross-filtering is required.

9. Current Findings

The figures below are transcribed from the current dashboard screenshots. They are useful for describing the current report, but should be labelled as current dashboard results until the underlying data rules and duplicate logic are fully validated.

Indicator	Current displayed value
Total Records	1,250
Records Requiring Review	319
Records Passing Checks	931
Review Rate	25.5%
Records Passing Checks	74.5%
Potential Duplicate Records	60
Missing Emails	87
Missing Phone Numbers	72
Invalid Emails	44
Missing Cities	33
Invalid Phone Numbers	26
Invalid Premium Values	16
Invalid Signup Dates	12
Invalid Dates of Birth	9

What the current figures suggest

- The dashboard flags 319 of 1,250 records for review, which is about 25.5% of the dataset.
- The missing email count (87) and missing phone count (72) indicate that contact information should be a focus for investigation.
- The 60 potential duplicate records should be checked against reliable identifiers before any records are merged or removed.
- The issue counts can overlap. One record may have more than one problem, so issue counts should not be added together to estimate unique records requiring review.

INTERPRETATION CAUTION
The 931 records passing checks are defined by the current model's flag logic. They should not be described as fully accurate or error-free until all quality rules and their coverage have been verified.

10. Corrective Actions and Recommendations

Issue	Recommended action	Priority shown
Missing Emails	Request missing email addresses and validate them before saving.	High
Missing Phone Numbers	Contact customers to obtain missing phone numbers.	Medium
Invalid Emails	Apply email-format validation and correct invalid addresses.	High
Invalid Phone Numbers	Standardise phone formats and validate the number of digits.	High
Missing Names	Make customer names mandatory during registration, with an exception process where needed.	High
Missing Cities	Request missing city information and update records.	Medium
Potential Duplicate Records	Compare reliable identifiers and merge only after confirming duplicates.	High
Invalid Signup Dates	Verify dates and correct confirmed errors using source information.	High
Invalid Dates of Birth	Verify dates of birth and apply sensible date-range checks.	High
Invalid Premium Values	Check premium amounts against approved business rules and correct confirmed errors.	High

Recommended ongoing controls

- Add validation at the point where customer information is entered.
- Use standard lists for policy types, statuses, and locations where appropriate.
- Assign an owner to investigate flagged records and record the outcome.
- Review recurring issues to identify where the process needs improvement.
- Refresh the report when the source data is updated and confirm that the results change as expected.

11. Validation, Testing, and Troubleshooting

The report has been checked for basic page navigation and selected visual interactions. The Executive Overview cross-filtering test worked when a policy type was selected. The Detailed Quality Analysis cards and corrective-action table displayed the expected current values and text.

Checks completed before final publication

- Confirm that each quality flag matches the intended rule and uses a consistent data type.
- Verify that the duplicate-count merge evaluates the complete dataset and that new records are included.
- Test the source refresh with a backup copy of the workbook and a controlled change; do not test on the only copy of the source file.
- Confirm that the total-record count and issue measures update correctly after refresh.
- Confirm the source file path and keep a safe backup of the original dataset.

Manual refresh workflow: update the Excel source while preserving its expected structure and location, save the workbook, open the Power BI project, and select Home → Refresh. If the source file is renamed or moved, the source path may need to be updated. Replacing the source file is only safe when the new workbook contains the intended complete dataset and has compatible columns.

12. Limitations and Assumptions

- The business and dataset are simulated for portfolio demonstration.
- The current counts reflect the dashboard at the time of documentation and may change after validation or refresh.
- Quality flags depend on the rules implemented in Power Query they do not replace manual review or business approval.
- Potential duplicate flags identify records to investigate, not confirmed duplicate customers.
- The project does not demonstrate compliance with every insurance, privacy, or data protection requirement. A separate legal and regulatory review would be needed for formal compliance conclusions.
- No automatic data correction is claimed; the dashboard identifies issues and recommends actions.

13. Conclusion

This project demonstrates how Excel, Power Query, DAX, and Power BI can be used together to create a practical view of customer data quality. The Executive Overview gives a quick summary of the dataset, while the Detailed Quality Analysis page helps users see specific issues and suggested actions.

The main value of the solution is that it turns a large set of customer records into a clearer list of issues to investigate. With final validation of the rules, duplicate logic, and refresh behaviour, the report can serve as a repeatable starting point for monitoring data quality and supporting better data governance.